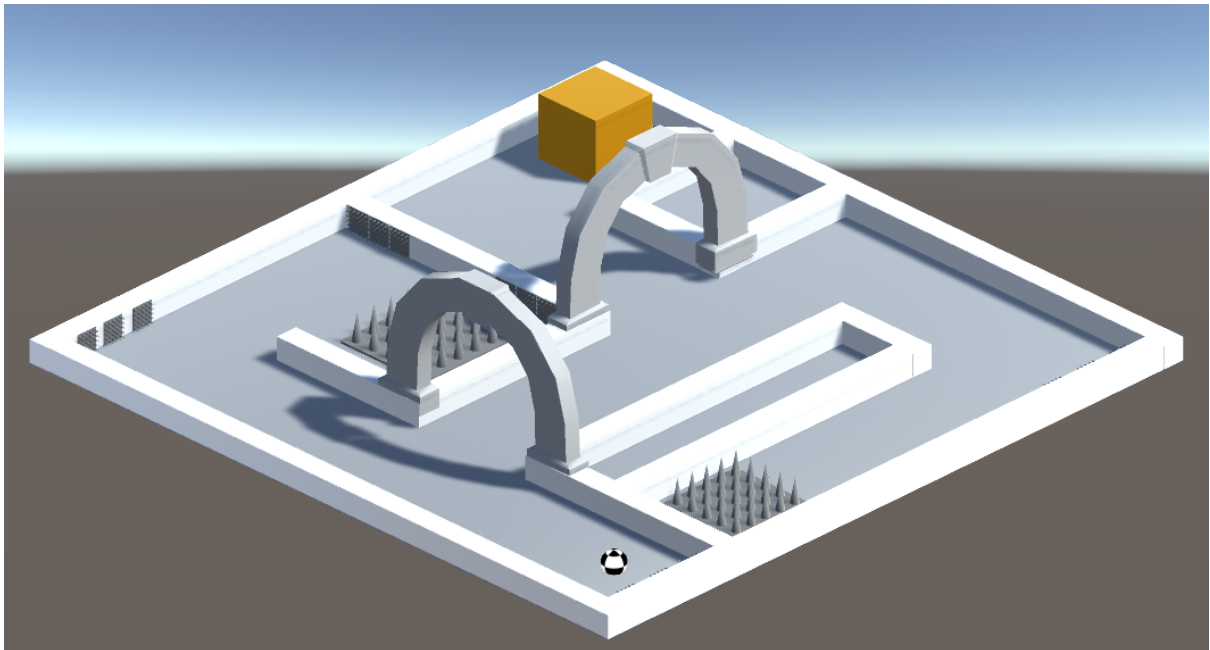


RollerBall Labyrinth Solver

In this project we have developed a game with ML Agents, the game consists of a labyrinth where a ball has to find a golden cube. The ball has to do so with some challenges: if the ball hits a trap it gets a negative reward and respawns, if it hits a wall, it also gets a punishment, but the game loop doesn't reset. In addition, the ball can't spend a lot of time in the maze, so we added a timer and if the ball hasn't solved the puzzle or died in that time the ball also dies. Finally, to avoid the ball staying still, we also check that the ball's speed isn't lower than a certain value. If so, we also give it a negative reward.

We have a lot of punishments, but how do we reward good behaviors? If the ball reaches the goal cube it gets the best positive reward, but to guide the ball we use a checkpoint system rewarding bit by bit through the path.

To make the process more visually appealing and easy to understand, the color of the floor changes depending on the ball's behavior. If the ball has died in the last episode, the floor turns red, if it wins, it turns green instead, but if it hits a wall in the current episode, it turns orange.



In the delivery, the project is already loaded with the best NN (RollerBall-7456689.onnx) in the RollerBallINN ID.

We also used various scenes at the same time for faster learning, but they are deactivated in the final release.

We attached two videos, one showing an agent working with the NN, and one to visualize the colors of the floor, because the ball didn't crash with the traps in the late stages of the training.